

Does written chain of thought keep substituting for the internal workspace as problems get harder?

J-space ablation on Qwen3-4B across GSM8K, MATH-500, and AIME

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1. Hypothesis

The question: when you erase a model’s verbalizable workspace, written chain of thought (CoT) is supposed to take over the work the workspace was doing. Does that substitution survive as problems get harder, or does it break down? The paper only checked the easy end (GSM8K, on Claude models). My bet: the substitution is partial. Writing externalizes the *storage* of intermediate results, but the planning between written tokens stays internal, and harder problems need more of it. Three predictions. P1: the paper’s GSM8K signature reproduces on Qwen3-4B, meaning direct answering collapses under J-space ablation while CoT mostly survives and a random control does little. P2, the primary one: the CoT-condition ablation drop grows with difficulty, faster than the random control’s. P3: ablation makes the model write more, re-deriving results it can no longer hold. The opposite outcomes were visible too: complete substitution means flat CoT drops matching the control; failure to transfer means J-ablation equal to random everywhere. One caveat on P1: I tuned ablation strength on a small pilot until that signature appeared, so P1 only validates the intervention. P2 and P3 ran on settings fixed before those cells; every later change to the plan is dated in the repo’s deviations log.

2. Experiment design

One checkpoint, Qwen/Qwen3-4B (hybrid thinking; revision 1cfa9a72). CoT vs direct is the `enable_thinking` template toggle, so both prompting conditions run identical weights. Grid: $\{\text{GSM8K, MATH-500, AIME 2024+2025}\} \times \{\text{CoT, direct}\} \times \{\text{clean, J-space ablation, random control}\}$. The ablation follows the paper: at every position, across a band of layers, zero the residual stream’s projection onto the span of the top- $k=10$ active J-lens vectors (rows of $W_U J_\ell$ with the final RMSNorm gain folded in), never ablating tokens in the top-10 of a clean pass. The paper does not say what the clean pass is during generation; I define it as the unablated model teacher-forced on the ablated stream’s own sampled tokens (two synced KV caches, $2 \times$ compute). The random control subtracts random directions at the same positions and layers, norm-matched per layer to the measured J-ablation; this is what separates “the J-space is special” from generic damage. A battery of automatic tasks (MMLU subset, one-step arithmetic, copy) and teacher-forced wikitext agreement check the other half of the paper’s claim. The direct condition prefills the assistant turn with `\boxed{}`: an instruction-only version leaked reasoning prose into 62% of “direct” answers, and the prefill cuts measured leakage to at most 3.4%. Difficulty is analyzed within MATH-500 (levels 1–5). The primary test: fit $\text{correct} \sim \text{ablated} \times \text{level}$ for J-vs-clean and for random-vs-clean, take D = the difference of the interaction coefficients, bootstrap clustered by problem; P2 is supported iff $D < 0$ with the 95% CI excluding zero. GSM8K and AIME serve as easy and hard anchors.

3. Experimental details

The J-lens is the Neuronpedia release for Qwen3-4B (fitted with Anthropic’s companion code on WikiText-103, 479 prompts, final-layer target, all 35 layers). My first acceptance check for it was miscalibrated: the lens agreed with the model’s top-1 at 70% of late-layer positions against a 75% bar, and a plain logit lens beats it on that metric. But the property the ablation needs is mid-stack readout of unspoken intermediates, and there the J-lens delivers what the logit lens cannot: on the boot-to-currency probe it reads the intermediate “Italy” at layers 23–26 and 28, where the logit lens shows nothing (three-way comparison, including my independently fitted lens, in `results/lens_compare.json`). The band came from Qwen’s own layer diagnostics (readout kurtosis, next-token accuracy, autocorrelation), not from the paper’s Sonnet band: layers 22–31 of 36, $k=10$, chosen on a 30-problem pilot. Thinking budgets 4096/8192/8192 tokens; the AIME cut from 16384 costs real accuracy (62.5% vs 33.3% on the same six problems), so AIME runs budget-censored. Scope by arm: GSM8K $n=200$ everywhere; MATH-500 all 500 clean and direct, with the ablated CoT arms on a seeded 30-per-level subsample ($n=150$) analyzed paired against the full clean cell; AIME 60 problems, 2 samples each except 1 for the ablated CoT arms. Sources: `openai/gsm8k` test; `HuggingFaceH4/MATH-500`; AIME = `Maxwell-Jia/AIME_2024 + math-ai/aime25`. One parser for every condition (math-verify), with parse failures counted separately from wrong answers. Compute: two A100-80GB pods, about 26 GPU-hours, $\approx \$43$. The README regenerates every number and figure from the committed results, raw transcripts included.

4. Experimental results

Three results. *First, the selective effect is real on this model.* CoT accuracy drops under J-ablation by +6.0 points on GSM8K (paired CI [+1.5, +11.0]) and +8.0 on MATH-500 ([+2.7, +14.0]), while the norm-matched random control moves -2.0 and -1.3 (CIs span zero). The automatic battery is untouched (MMLU 66/66/67, arithmetic 100/100/99, copy 100/100/100 for clean/J/random) though J-ablation shifts 31% of wikitext next-token choices (random: 7%). The strict direct condition starts at 24.0/22.2/0.0% on GSM8K/MATH/AIME and barely reacts to anything; its largest J-

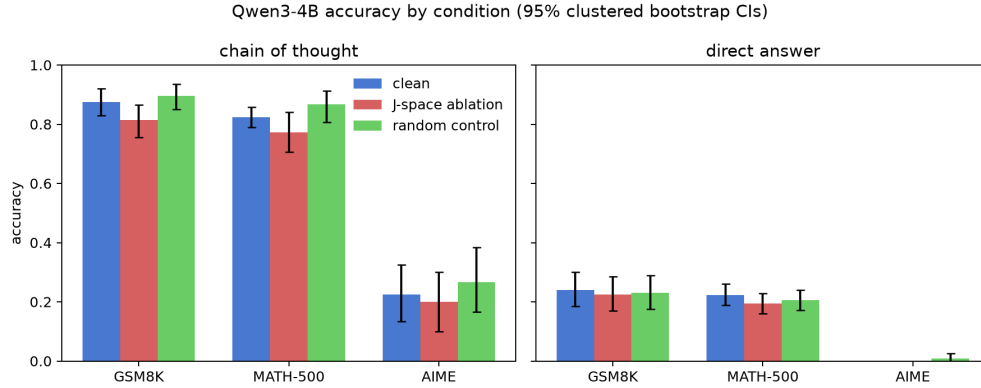


Figure 1: Accuracy by condition (95% clustered bootstrap CIs). In the CoT panel the J-bar sits below clean and control on all three datasets; in the direct panel the ~22-24% floor barely moves.

effect is +2.8 points on MATH-500 ($[+0.4, +5.4]$). *Second, the primary test is negative:* $D = +0.42$, CI $[-0.13, +1.27]$, so P2 is not supported; the J-cost is difficulty-flat on both the logit and probability scales (Fig. 2). P3 is also wrong: ablated CoT is *shorter* (median 3146 vs 3689 tokens on MATH), slightly more repetitive (repeated-8-gram rate 0.056 vs 0.037), and, only under J-ablation, the model sometimes ends its turn with no answer at all (8% on GSM8K; 0% in clean and random). *Third, reading those transcripts explains the flat trend.* They contain complete, usually correct solutions inside the think block: the model solves the problem, then fails the close-think-and-answer step. Re-grading leniently (accept a boxed answer anywhere) makes the GSM8K J-drop vanish (-0.5 , CI $[-3.5, +2.5]$) while the MATH-500 drop persists at $+9.3$ ($[+4.0, +14.7]$) with random at exactly 0.0. On AIME every completed run is correct in every arm; completion rates are 22.5/20.0/26.7% for clean/J/random (n of 120/60/60), pointing the same way as MATH but underpowered. Dose-response on GSM8K: clean 87.5, $k=5$: 85.5, $k=10$: 81.5, $k=25$: 82.5.

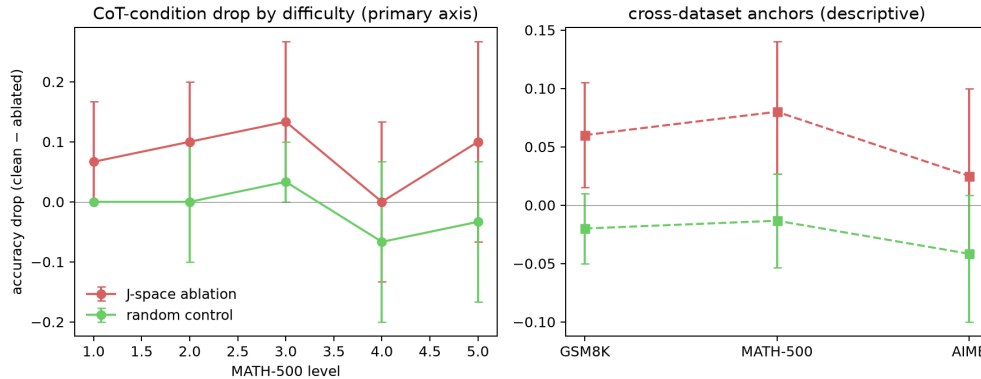


Figure 2: CoT-condition ablation drop, J vs random, by MATH level (left) and across datasets (right). The J-line sits above the control everywhere; the trend is flat.

5. Analysis of results

The main finding: erasing the J-space reliably breaks the model’s ability to *deliver* an answer it has already worked out on the page, and only once problems are hard enough does it also break the working-out itself; written CoT fully protects the computation on easy problems and only partly on hard ones. My original prediction fails as stated, because I predicted the wrong observable: the strict accuracy drop is flat with difficulty, but that flatness is two opposite trends canceling. The delivery disruption is difficulty-flat and dominates GSM8K, where the math inside the think block survives ablation essentially intact. The reasoning cost appears from MATH-500 up ($+9.3$ under lenient grading) and matches the storage-vs-manipulation intuition I started from, though I hold that version more weakly: the lenient re-grade is exploratory, chosen after reading transcripts. Alternatives I cannot rule out: lenient grading may credit answers the model would not have committed to (the random arm shows zero such inflation, which reassures me but does not settle it); MATH levels 4–5 are budget-censored in both arms (truncation 20–43%); AIME rests on 60 problems; and the band, k , and the strict no-externalization operationalization are choices that shape effect sizes. That last choice cuts the other way too: held at ~22–24% when forbidden from writing, Qwen3-4B seems to do most of its apparent direct answering by externalized micro-reasoning, so the paper’s direct-mode collapse may partly measure verbalization rather than reasoning. With more compute: AIME at 16k with pass@ k , think-phase vs answer-phase ablation to isolate the delivery effect, and more seeds.